

Experiment 21: Image Sharpening Using Laplacian Mask with Diagonal Neighbors

Aim

To perform **image sharpening using a Laplacian mask with an extension of diagonal neighbors** and enhance the edges and fine details of an image.

Laplacian Mask

The given mask is:

1	1	1
1	-8	1
1	1	1

This mask uses **all 8 neighboring pixels**, including the diagonal neighbors.

Procedure

1. Read the input image.
2. Convert the image into grayscale.
3. Apply the Laplacian mask to the grayscale image.
4. Since the center coefficient is **-8**, subtract the Laplacian result from the original image.
5. Normalize the pixel values to the range 0–255.
6. Display the original and sharpened images.
7. Save the sharpened image.

Python Program

```
import cv2
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt

# Read input image

img = cv2.imread("input.jpg")

if img is None:

    print("Error: Image not found. Check the file name and path.")

    exit()

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Laplacian mask with diagonal neighbors

mask = np.array([

    [1, 1, 1],

    [1, -8, 1],

    [1, 1, 1]

], dtype=np.float32)

# Apply Laplacian mask

laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)

# Sharpen image

sharpened = gray.astype(np.float32) - laplacian
```

```
# Convert values to 0-255

sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)


# Display images

plt.figure(figsize=(10, 4))


plt.subplot(1, 2, 1)

plt.imshow(gray, cmap="gray")

plt.title("Original Image")

plt.axis("off")


plt.subplot(1, 2, 2)

plt.imshow(sharpened, cmap="gray")

plt.title("Laplacian Sharpened Image")

plt.axis("off")


plt.show()


# Save output

cv2.imwrite("sharpened_output.jpg", sharpened)
```

Input



Output

Original Image



Laplacian Sharpened Image



Result

The image was successfully sharpened using the **3×3 Laplacian mask with diagonal neighbors**. The edges and fine details of the input image are enhanced.